

Deep Learning

AI302

Books

- Deep Learning. Book by Ian Goodfellow and Yoshua Bengio and Aaron Courville, The MIT
- Dive into Deep Learning: Book by Aston Zhang, Zack C. Lipton, Mu Li, Alex J. Smola.
- Charu C. Aggarwal. Neural Networks and Deep Learning: A Textbook. Springer. 2019.
- NPTEL/YouTube lecture series on Deep Learning by Mitesh M. Khapra (IIT Madras)

Syllabus

- **Introduction to Deep learning:** Overview of Applications, Artificial Neural Networks (ANNs), Activation Functions, Forward and Backpropagation Algorithms, Gradient Descent, Optimization Algorithms
- **Modern Convolution Neural Networks:** Deep Convolution Neural Networks (AlexNet), Network using Blocks (VGG), Network in Network (NiN), Multi-Branch Networks (GoogLeNet), Residual Networks (ResNet), Densely Connected Network (DenseNet), TransferLearning and Fine-Tuning
- **Modern Recurrent Neural Networks:** Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Deep Recurrent Neural Networks, Bidirectional Recurrent Neural Networks, Variational Auto encoders (VAE), Generative Adversarial Networks (GANs)

Syllabus

- **Applications of GANs and Advanced Topics:** Image-to-Image Translation (pix2pix), Neural Style Transfer (Style GAN), future of Generative Modeling, Explainable AI (XAI)
- **Introduction to Transformers and its Applications:** Attention Mechanisms, Natural Language, Transformer Language Models, Sequence-to-Sequence transformers, Vision Transformers, BERT, GPT

Course Evaluation Scheme

Theory Component (3 Credits)

Total Weightage: 100 Marks

- **Mid-Semester Examination:** 30 Marks
- **Quizzes:** 20 Marks
- **End-Semester Examination:** 50 Marks

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Laboratory Component (1 Credit)

Total Weightage: 50 Marks

- **Continuous Evaluation:** 25 Marks
- **External Examination/Project:** 25 Marks

Acknowledgment

I sincerely thank **Dr. Mitesh M. Khapra** from **IIT Madras** for his slides, most of which I refer to in this course.

Deep learning

- Deep learning is a subset of machine learning that focuses on training neural networks with many layers (hence “deep”) to to analyze and learn from data
- Not all ML methods/algorithm are deep learning (e.g., SVM, decision trees, k-means)

History of Deep Learning

1943:

- In 1943, Warren McCulloch and Walter Pitts presented a mathematical model of a biological neuron, known as the **McCulloch–Pitts neuron**.
- McCulloch Pitts Neuron has very limited capability and has no learning mechanism
- Despite its limitations it will lay the foundation for artificial neural network & deep learning.

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1957:

- In 1957, Frank Rosenblatt introduced an improved version of the McCulloch–Pitts neuron called **the Perceptron**.
- The perceptron could learn from data and perform binary classification on its own.
- This breakthrough inspired many years of research on shallow neural networks.

History of Deep Learning

1965: Birth of Multilayer Neural Network

- In 1965, Alexey Ivakhnenko, along with Valentin Lapa, developed a hierarchical neural network that used polynomial activation functions and was trained with the Group Method of Data Handling (GMDH).
- This is considered the **first multi-layer perceptron**, and Ivakhnenko is often called the father of deep learning.

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1969: The Fall of Perceptron

- In 1969, Marvin Minsky and Seymour shows that a single perceptron **cannot solve** complex problems like the **XOR function**. To solve such problems, perceptron's need **multiple hidden layers**, but this makes the original perceptron learning algorithm ineffective.

History of Deep Learning

1970: Backpropagation is computer coded

- Seppo Linnainmaa published a **general method for automatic differentiation**, which laid the foundation for **backpropagation**
- Although backpropagation is now widely used, it **was not implemented in neural networks till next decade.**

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1980: First CNN Architecture

- Kunihiro Fukushima comes up with **Neocognitron**, the first convolutional neural network (**CNN**) architecture which could recognize visual patterns such as handwritten characters.

History of Deep Learning

1980: Hopfield Network – Early RNN

- In 1980, John Hopfield created the **Hopfield Network**, a type of **recurrent neural network (RNN)**.
- It works as a **content-addressable memory system** and later inspired many modern RNN models in deep learning.

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1982: Proposal for Backpropagation in ANN

- Paul Werbos, in his 1974 Ph.D. thesis, proposed the **use of Backpropagation** for **propagating errors** during the training of Neural Networks.
- His results of the Ph.D. thesis will eventually lead to the **practical adoption of backpropagation** by the neural network community in the future.

History of Deep Learning

1985: Boltzmann Machine

- David H. Ackley, Geoffrey Hinton and Terrence Sejnowski create Boltzmann Machine that is a **stochastic recurrent neural network**.
- This neural network has only input layer and hidden layer **but no output layer**.

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1986: Implementation of Backpropagation

- Geoffrey Hinton, David Rumelhart, and Ronald Williams successfully **implemented backpropagation in neural networks**.
- It opened gates for training complex deep neural network, overcoming a major obstacle in earlier research.

History of Deep Learning

1989: Boltzmann Machine

- Yann LeCun used **backpropagation** to train a **convolutional neural network (CNN)** to recognize **handwritten digits**.
- This was a breakthrough, laying the foundation for modern computer vision with deep learning.
- In 1989, George Cybenko proved that a **feedforward neural network with a single hidden layer** and a finite number of neurons can approximate any continuous function, giving strong support to the potential of deep learning.

History of Deep Learning

2006: GPU Revolution Begins

- Andrew NG's group in Stanford starts **encouraging the use of GPUs** to train deep neural networks to speed up the training time by several times.
- This made it practical to train deep learning models on **large volumes of data efficiently**.

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2009:

- In 2009, Fei-Fei Li created the **ImageNet**, a **database** (with 14 million labeled images), providing a large-scale dataset that significantly advanced deep learning in computer vision.
- It became an important **benchmark** for deep learning researchers, who would participate in ImageNet competitions (ILSVRC) every year.

History of Deep Learning

2011:

- Yoshua Bengio, Antoine Bordes, Xavier Glorot shows that ReLU activation function can **avoid vanishing gradient problem**.
- That is, apart from GPU, deep learning community has another tool to avoid issues of longer and impractical training times of deep neural network.

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2012:

- In 2012, Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton won the ImageNet competition using **AlexNet**, a deep CNN, with 84% accuracy, a big jump from the previous 75%

History of Deep Learning

2014: The birth of GANs

- In 2014, Ian Goodfellow introduced **Generative Adversarial Networks (GANs)**, **which** opened up new doors of applications in fashion, art, and science because they can generate realistic synthetic data.

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2015: AlphaGo beats human

- In 2015, Google DeepMind's AlphaGo (deep reinforcement learning model) **beats the human champion** in the complex game of Go, showcasing deep learning's ability to solve complex strategy problems.
- Since the game is much more complex than chess, this shows deep learning's ability to solve complex strategy problems.

History of Deep Learning

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- In 2016, deep learning frameworks such as **TensorFlow and PyTorch** emerged, making deep learning more accessible to researchers and practitioners.

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2020:

- In 2020, OpenAI **introduced GPT-3**, a large language model with 175 billion parameters, demonstrating the strong capabilities of deep learning in natural language processing.

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Ongoing research: Deep learning continues to evolve with advances in reinforcement learning, unsupervised learning, and multimodal learning.

1943: McCulloch-Pitts Neuron Model

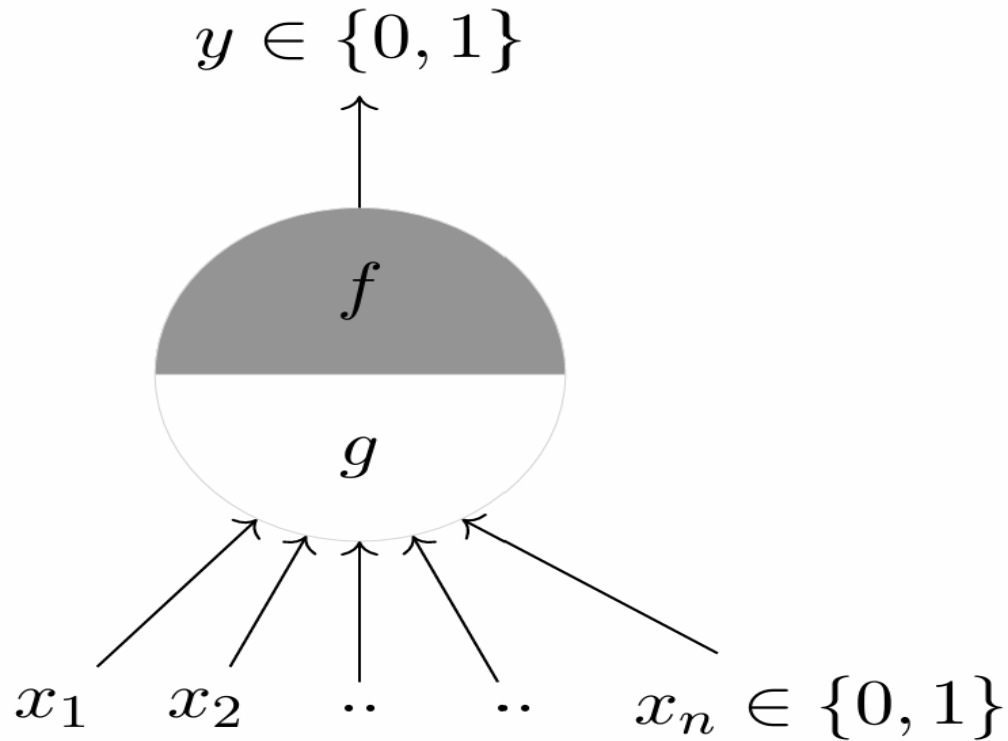
- McCulloch– Pitts Neuron Model was formulated in the year by Warren McCulloch (neuroscientist) and Walter Pitts ((logician)) in the year 1943
- It is built on a clear, well-structured mathematical foundation.
- The neuron allows only the binary states i.e., ‘0’s and ‘1’s, so it is called as a binary activated neuron.
- These neurons are connected by direct weighted path. The connected path can be excitatory or inhibitory, i.e.; each input can be excitatory or inhibitory

1943: McCulloch-Pitts Neuron Model

- The McCulloch-Pitts Neuron is the first computational model of a neuron. It can be divided into two parts:
 - **Aggregation:** The neuron aggregates multiple boolean inputs (0 or 1).
 - **Threshold Decision:** Based on the aggregated value, the neuron makes a decision using a threshold function.

Note: The neuron fires (outputs 1) if the aggregated sum of inputs meets or exceeds a threshold value (θ).

1943: McCulloch-Pitts Neuron Model



- g aggregates the inputs and the function f takes a decision based on this aggregation
- The inputs can be excitatory or inhibitory

$$g(x_1, x_2, \dots, x_n) = g(\mathbf{x}) = \sum_{i=1}^n x_i$$

$$y = f(g(\mathbf{x})) = \begin{cases} 1 & \text{if } g(\mathbf{x}) \geq \theta \\ 0 & \text{if } g(\mathbf{x}) < \theta \end{cases}$$

- θ is called the thresholding parameter

1943: McCulloch-Pitts Neuron Model

Example: Wanting to predict whether to watch a football game. The inputs (boolean values) could be:

X1: Is Premier League on? (1 if yes, 0 if no)

X2: Is it a friendly game? (1 if yes, 0 if no)

X3: Are you not at home? (1 if yes, 0 if no)

X4: Is Manchester United playing? (1 if yes, 0 if no)

If you always watch the game when at least two conditions are met, **θ would be 2.**

Here **X3 is inhibitory** because you can't watch the game at home.

McCulloch-Pitts Neuron representing various boolean functions:

AND Function: Fires when all inputs are ON ($x_1 + x_2 + x_3 \geq 3$).

OR Function: Fires when any input is ON ($x_1 + x_2 + x_3 \geq 1$).

Inhibitory Input Function: Fires only when specific conditions are met (e.g., x_1 AND NOT x_2).

NOR Function: Fires when all inputs are OFF.

NOT Function: Inverts the input.

Thank You